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6.1.1. Incremented Date

Peusdocode:

1. START

2. READ three integers from the user representing the day, month, and year, and STORE them in variables d, m, and y.

3. INITIALIZE a boolean variable flag to True to keep track of date validity.

4. CHECK basic validity: IF y <= 0 OR m < 1 OR m > 12, SET flag to False.

5. IF flag is True:

    a. DETERMINE maxdays based on the month:

6. IF m is 1, 3, 5, 7, 8, 10, or 12: maxdays = 31.
7. ELSE IF m is 4, 6, 9, or 11: maxdays = 30.
8. ELSE IF m is 2 (February):
9. IF the year y is a leap year ((y % 400 == 0) or (y % 4 == 0 and y % 100 != 0)): maxdays = 29.
10. ■ELSE: maxdays = 28.
11. ○ CHECK day validity: IF d < 1 OR d > maxdays, SET flag to False.

12. IF flag is True (The date is completely valid): ○ INCREMENT the day d by 1.

    a. IF d > maxdays (End of the month reached):

13. SET d = 1.
14. INCREMENT the month m by 1.
15. IF m > 12 (End of the year reached):
16. SET m = 1.
17. INCREMENT the year y by 1.

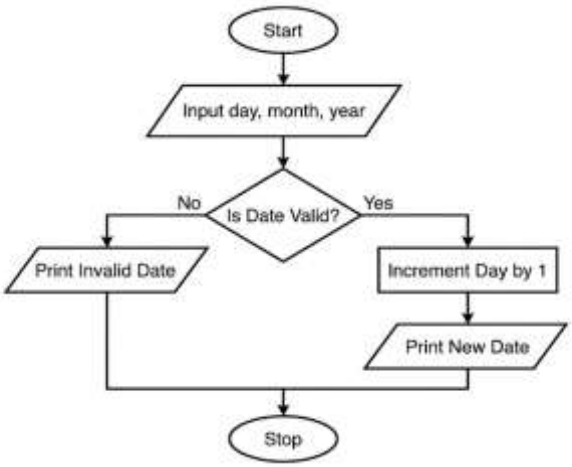
18. PRINT the incremented date formatted as DD-MM-YYYY (zero-padded).

19. ELSE:

    a. PRINT "Invalid Date".

20. END

Flowchart:



Code:

```
d = int(input())
m = int(input())
y = int(input())
flag = True
maxdays = 0

if y <= 0 or m < 1 or m > 12:
    flag = False
 if flag:
    if m in (1, 3, 5, 7, 8, 10, 12):
            maxdays = 31
    elif m in (4, 6, 9, 11):
            maxdays = 30
    elif m == 2:
            if (y % 400 == 0) or (y % 4 == 0 and y % 100 != 0):
                    maxdays = 29
            else:
                    maxdays = 28
    if d < 1 or d > maxdays:
            flag = False
 if flag:
    d += 1
    if d > maxdays:
            d = 1
            m += 1
            if m > 12:
                    m = 1
                    y += 1

    print(f"{d:02d}-{m:02d}-{y:04d}")
else:
    print("Invalid Date")
```